

Rambarki Pavan Kumar

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Professional Summary

Detail-oriented VLSI Engineer and NIT Rourkela M.Tech graduate with hands-on experience in UVM-based verification, AMBA protocol debug, and EDA tool flows (Synopsys VCS, Siemens QuestaSIM, Cadence Virtuoso, Xilinx Vivado). Strong foundation in RTL design, digital circuits, and SoC architecture, with additional expertise in AI/ML frameworks (PyTorch) and a keen interest in leveraging AI tools to automate verification workflows.

Technical Skills

- **Hardware Description/Verification:** SystemVerilog, UVM, Verilog HDL
- **Verification Concepts:** UVM Phases & Objections, Sequences & Virtual Sequences, Driver, Monitor, Scoreboards, Verification Intellectual Property (VIP), Accelerated Verification Intellectual Property (AVIP), Functional Coverage, Constrained Random Verification, SystemVerilog Assertions (SVA), RAL (Frontdoor/Backdoor), Waveform & Log Debugging.
- **Protocols & Hardware:** AMBA AXI4, AXI4-Lite, APB, CHI, RISC-V Architecture, FIFO, CDC (Clock Domain Crossing)
- **EDA Tools:** Synopsys VCS, Siemens QuestaSIM, Xilinx Vivado, Cadence Virtuoso, LTSpice, TSpice, LabVIEW
- **Programming & Frameworks:** C, Python, PyTorch, Jupyter Notebook
- **Operating Systems:** Linux, Windows

Professional Experience

Verification Engineer

July 2025 - Present

Miraфра Technologies, Bengaluru

- **UVM Environment Development:** Designing and enhancing a UVM-based DMA verification environment, leveraging Register Abstraction Layer (RAL) for strict control and status register validation.
- **Protocol Analysis & Debugging:** Analyzing AMBA AXI4-Lite protocol transactions, focusing on register-level handshakes, data movement efficiency, and interconnect utilization.
- **Root-Cause Analysis:** Identifying functional anomalies and performance bottlenecks across diverse hardware blocks by deep-diving into UVM transaction logs and waveform tools (VCS/QuestaSIM).
- **Collaborative Verification:** Reviewing RTL specs to build comprehensive test plans and collaborating closely with design teams to resolve protocol violations.

Teaching Assistant

May 2024 - May 2025

Department of Electronics & Communication Engineering, NIT Rourkela

- Conducted Instrumentation Laboratory sessions, demonstrating hands-on usage of oscilloscopes, DSOs, multimeters, P/PI/PID controllers, process controllers, and biomedical instrumentation equipment to undergraduate students.
- Evaluated student lab performance and provided technical support for troubleshooting experimental setups.

Technical Projects

UVM-Based DMA Verification with RAL

- Developed reusable UVM testbench consisting of driver, monitor, sequencer, agent, environment, scoreboard and virtual sequences.
- Integrated UVM Register Abstraction Layer (RAL) supporting frontdoor/backdoor access.
- Developed functional coverage and assertions for register verification.
- Performed register prediction and mirror checking.

AXI4 Protocol Debug & Compliance Project

- Tested and verified AXI4 read/write transactions at the signal level, isolating protocol bugs by tracking channel handshakes (AW, W, B, AR, R).
- Evaluated complex transaction scenarios, including variant burst types, address alignment configurations, ID tags, and out-of-order responses.
- Developed SystemVerilog Assertions (SVA) to capture READY/VALID backpressure issues and ensure absolute compliance with AMBA specifications.

Synchronous and Asynchronous FIFO Design

- Designed robust Synchronous and Asynchronous FIFO buffers in Verilog to bridge distinct clock domains safely.
- Mitigated Clock Domain Crossing (CDC) metastability hazards by implementing Gray-coded pointers and multi-flop synchronizers.
- Authored self-checking testbenches in Vivado to validate corner cases including simultaneous read/write, reset states, overflow, and underflow.

Design of Two-Stage Operational Amplifier

- Designed a high-gain, stable two-stage CMOS operational amplifier using a 180nm process node within Cadence Virtuoso.
- Optimized phase margin and loop stability under varying capacitive loads utilizing Miller compensation networks.
- Executed DC, AC, and transient analyses to evaluate gain-bandwidth product (GBW), slew rate, and power dissipation parameters.

Advanced Peripheral Bus (APB) Slave Interface Design

- Implemented an AMBA APB slave interface in Verilog utilizing an explicit Finite State Machine (FSM) to handle setup and access phases.
- Verified flawless signal execution across control paths using PADDR, PSEL, PENABLE, PWRITE, and PREADY.

Brain tumor Segmentation using Deep Learning Techniques

- Developed a novel attention-based deep learning framework for brain tumor segmentation using single-channel MRI slices from the BraTS 2020 dataset.
- Performed data pre-processing, including normalization to optimize model input quality.

Education

M.Tech in Electronics and Instrumentation Engineering

National Institute of Technology, Rourkela

Aug 2023 – May 2025

CGPA: 8.47/10

- **Core Coursework:** CMOS Digital IC Design, Analog & Mixed Signal VLSI, Hardware Modelling using Verilog HDL, Semiconductor Physics, Static Timing Analysis, Machine Learning, Computer Vision.

B.Tech in Electronics and Communication Engineering

Guru Ghasidas Vishwavidyalaya (A Central University), Bilaspur

Aug 2019 – May 2023

CGPA: 7.96/10

- **Core Coursework:** Digital Electronics, Analog Electronics, Digital Signal Processing, Artificial Intelligence, Signals and Systems

Publications & Book Chapters

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- **Conference Paper:** Rambarki Pavan Kumar, Pranshu Jena, Umesh C. Pati, "*Effects of Novel Segmentation Framework ConvUNext Network along with Spatial Attention Module on Small Brain Tumor Dataset*", IEEE IC2E3 2025, NIT Uttarakhand. [**Published**]
 - **Conference Paper:** Rambarki Pavan Kumar, Pranshu Jena, Umesh C. Pati, "*RegUNeXt with Hybrid Attention for Advanced Segmentation for Tumorous Region in Brain Tumor*", 4th IEEE CVMI 2025, NIT Rourkela. [**Published**]
 - **Book Chapter:** Rambarki Pavan Kumar, Pranshu Jena, Umesh C. Pati, "*Artificial Intelligence in Medical Imaging*", in *Intelligence-driven Automation Techniques*, IEEE Press–Wiley. [**Accepted**]

Achievements & Awards

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- **Best Paper Award** - International Conference on Communication, Electronics and Electrical Engineering (IC2E3-2025).
 - **VLSI Industry Certification** - VLSI Certification in collaboration with 1Stop IIT Guwahati